PRODUCT LINE PROFITABILITY & MARGIN PERFORMANCE ANALYSIS FOR NASSAU CANDY DISTRIBUTOR

Internship Project Report Submitted by Shrushti Bhambare

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UNIFIED MENTOR PVT. LTD.

Gurugram,Haryana

CERTIFICATE

This is to certify that **Shrushti Bhambare** has undertaken the Data Science Internship project titled **"Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor"** with Unified Mentor Pvt. Ltd.

The project focuses on analyzing sales transactions, product costs, gross profit, and profit margins to understand the financial performance of different products and divisions. The work involves data cleaning, exploratory data analysis, profitability calculations, data visualization, and the development of an interactive dashboard.

The project utilizes Python and its data analysis libraries, including Pandas and NumPy, along with Plotly for visualization and Streamlit for dashboard development. The analysis aims to identify high-profit products, evaluate low-margin products, and generate meaningful business insights to support data-driven decision-making.

This report presents the work carried out as part of the internship project.

Project Title: Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Submitted by: Shrushti Bhambare

Organization: Unified Mentor Pvt. Ltd.

Acknowledgement

I would like to express my sincere gratitude to **Unified Mentor Pvt. Ltd.** for providing me with the opportunity to undertake this Data Science internship project titled *"Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor."*

I am thankful to my internship mentor for the guidance, support, and valuable feedback provided throughout the project. This opportunity helped me understand the practical applications of data science, exploratory data analysis, data visualization, and business analytics.

I would also like to acknowledge the developers of Python and its associated libraries, including Pandas, NumPy, Plotly, and Streamlit, which supported the data analysis and dashboard development process.

I extend my gratitude to my college faculty for their academic guidance and encouragement. Finally, I am thankful to my family and friends for their continuous motivation and support throughout the completion of this project.

This project has been a valuable learning experience, allowing me to apply theoretical concepts to a practical business problem and develop a better understanding of data-driven decision-making.

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Abstract

Profitability analysis is essential for businesses to understand their financial performance, identify profitable products, and make informed decisions regarding pricing and resource allocation. This study presents a Product Line Profitability and Margin Performance Analysis for Nassau Candy Distributor using transactional sales data.

The project uses Python, Pandas, NumPy, Plotly, and Streamlit to perform data cleaning, exploratory data analysis, profitability calculations, and interactive visualization. Key performance indicators such as total sales, total cost, gross profit, gross profit margin, and units sold are analyzed. The study also compares product divisions, evaluates individual product profitability, and identifies products with comparatively low profit margins.

The analyzed dataset contains 10,194 records and 18 columns. The analysis reports total sales of $141,783.63, total cost of $48,340.83, and gross profit of $93,442.80, resulting in an overall gross profit margin of approximately 65.91%. The Chocolate division contributes the largest share of sales and gross profit. Product-level analysis also reveals differences in profitability, highlighting the importance of evaluating both sales contribution and margin performance.

An interactive Streamlit dashboard is developed to present financial indicators, product comparisons, margin analysis, and profitability trends in an accessible format. The findings can support product portfolio evaluation, pricing reviews, and data-driven business planning.

Keywords: Profitability Analysis, Gross Profit Margin, Sales Analytics, Product Performance, Business Intelligence, Exploratory Data Analysis, Data Visualization, Streamlit.

Executive Summary

The project titled **"Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor"** focuses on evaluating product-level and division-level financial performance using sales transaction data. The primary purpose is to understand the relationship between sales revenue, product costs, gross profit, and gross profit margins.

The project uses Python, Pandas, NumPy, Plotly, and Streamlit for data preprocessing, exploratory data analysis, financial calculations, visualization, and interactive dashboard development. The dataset contains 10,194 records and 18 columns related to product sales, costs, divisions, regions, and other transaction attributes.

The analysis reports total sales of **$141,783.63**, total cost of **$48,340.83**, and gross profit of **$93,442.80**. The overall gross profit margin is approximately **65.91%**. These indicators provide an overview of the financial performance represented in the dataset.

Division-level analysis shows that the Chocolate division contributes the largest share of sales and gross profit. Product-level analysis identifies Wonka Bar – Scrumdiddlyumptious as the highest gross-profit product, generating $19,357.50 in gross profit with a margin of approximately 69.44%. In contrast, Kazookles records a comparatively low gross profit margin of approximately 7.69%, highlighting the importance of evaluating product profitability beyond sales revenue alone.

An interactive Streamlit dashboard is developed to present key performance indicators, product comparisons, division performance, cost and margin analysis, and profit concentration. The dashboard allows users to explore profitability information through interactive charts and filters.

The findings demonstrate how data analysis and visualization can help identify profitable products, recognize low-margin items, and support product portfolio evaluation. The project also establishes a foundation for future improvements, such as pricing analysis, detailed cost investigation, and profitability forecasting, subject to the availability of suitable data.

In conclusion, the project demonstrates the practical application of data science and business analytics in understanding product profitability and supporting informed business decisions.

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Introduction

In a competitive business environment, analyzing sales and profitability is important for understanding financial performance and supporting effective decision-making. Sales figures alone do not provide a complete picture of business success because products with high sales may not always generate the highest profit margins.

Nassau Candy Distributor deals with multiple products across different divisions. Evaluating the performance of these products requires an understanding of sales revenue, product costs, gross profit, and profit margins. Such analysis can help identify profitable products, recognize low-margin items, and compare the financial contribution of different product divisions.

This project applies data analysis and visualization techniques to examine the profitability performance of Nassau Candy Distributor. Python and its data analysis libraries are used to clean and process the dataset, calculate financial indicators, and conduct exploratory data analysis. Interactive visualizations are developed using Plotly and Streamlit to present the results in a user-friendly dashboard.

The study focuses on product-level and division-level profitability, cost comparisons, gross margin performance, and profit concentration. The resulting insights can assist business stakeholders in reviewing product performance and identifying opportunities for improving profitability.

Problem Statement

Businesses managing multiple products need to understand which products generate the highest profits and which products have comparatively low profit margins. Analyzing sales revenue alone may lead to incomplete conclusions because product costs and margins also influence overall financial performance.

The project addresses the following problems:

- Identifying the overall sales, cost, and gross profit performance of Nassau Candy Distributor.
- Comparing profitability across different product divisions.
- Identifying high-profit and low-margin products.
- Examining the relationship between sales revenue, product cost, and gross profit.
- Understanding the concentration of profit among individual products.
- Presenting the results through an interactive dashboard for easier business analysis.

The project uses data-driven techniques to convert transactional records into meaningful profitability indicators and business insights.

Objectives

The primary objectives of this project are:

1. To analyze the overall sales and financial performance of Nassau Candy Distributor.
2. To calculate total sales, total cost, gross profit, and gross profit margin.
3. To compare profitability across product divisions.
4. To identify products that generate the highest gross profit.
5. To identify products with comparatively low gross profit margins.
6. To analyze product cost and revenue relationships.
7. To examine the concentration of gross profit among products.
8. To develop an interactive Streamlit dashboard for profitability analysis.
9. To derive business insights that may support product portfolio evaluation and pricing decisions.

Literature Review

Product profitability and margin analysis have become important areas in business intelligence because sales and cost data help businesses understand their financial performance, identify profitable products, and make better business decisions. Analyzing past business data helps organizations improve product performance and increase profitability.

Previous studies have highlighted the importance of cost analysis and profitability measurement. Comparing product sales and costs helps businesses calculate gross profit and profit margin. This analysis allows organizations to identify products that generate higher profits and products that may need improvement.

Product-level analysis helps businesses compare the performance of different products and divisions. By analyzing sales, costs, gross profit, and gross profit margin, businesses can understand which products contribute the most to overall profit and which products have lower profitability.

Exploratory Data Analysis (EDA) and data visualization play an important role in understanding business data. Using tools such as Python, Pandas, NumPy, and Plotly, large datasets can be analyzed and presented through charts, graphs, and statistical summaries. These methods make it easier to identify patterns and compare product performance.

Interactive dashboards developed using Streamlit help present important business information through charts, filters, and key performance indicators (KPIs). They allow users to explore product profitability and compare business performance without directly working with raw data or code.

Therefore, product line profitability analysis is useful for businesses such as Nassau Candy Distributor. Identifying high-profit products helps businesses understand their major sources of profit, while identifying low-margin products helps them recognize areas for improvement. This analysis supports better pricing decisions, cost management, and overall business performance.

Tools & Technologies

The project Product Line Profitability and Margin Performance Analysis for Nassau Candy Distributor uses the following tools and technologies to analyze the dataset and develop an interactive dashboard.

Python: Used as the main programming language for data analysis and calculations.

Pandas: Used to load, clean, organize, and analyze the dataset.

NumPy: Used for numerical calculations and data processing.

Plotly: Used to create interactive charts and graphs for product profitability analysis.

Streamlit: Used to develop an interactive dashboard to display key performance indicators (KPIs), product comparisons, and business insights.

Jupyter Notebook: Used for Exploratory Data Analysis (EDA), data cleaning, and visualization.

CSV Dataset: The Nassau Candy Distributor dataset is used to analyze product sales, costs, gross profit, and profit margins.

Dataset Description

The dataset Nassau Candy Distributor.csv contains 10,194 rows and 18 columns.

Column Name	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date of the order
Ship Date	Date of shipment
Ship Mode	Shipping method
Customer ID	Unique customer identifier
Customer Name	Name of the customer
Country/Region	Country or region of the customer
City	Customer's city
State/Province	Customer's state or province
Postal Code	Customer's postal code
Division	Product division
Region	Business region
Product ID	Unique product identifier
Product Name	Name of the product
Sales	Revenue generated from sales
Units	Number of units sold
Gross Profit	Profit earned after deducting product cost

Data Cleaning & Validation

1. **Load Dataset:** Imported the CSV file using Pandas.
2. **Check Dataset Structure:** Reviewed the number of rows, columns, and column names.
3. **Check Missing Values**: Checked all columns for missing values. No missing values were found.
4. **Check Duplicates:** Checked for duplicate records. No duplicate rows were found.
5. **Validate Row IDs:** Checked for duplicate Row IDs.
6. **Check Data Types:** Reviewed data types and ensured date and numerical columns were suitable for analysis.
7. **Validate Numerical Values:** Checked sales, units, and profit values for consistency.
8. **Verify Profit Calculations:** Compared gross profit with sales minus cost, where cost data was available.
9. **Prepare Data for Analysis:** Used the validated dataset for EDA, visualization, and dashboard development.

Methodology

Dataset Collection

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Data Loading

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Data Cleaning and Validation

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Exploratory Data Analysis (EDA)

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Profitability Calculation

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Data Visualization

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Streamlit Dashboard Development

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Results and Business Insights

Exploratory Data Analysis

Dataset Preview

Dataset shape: (10194, 18)

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Country/Region	City	State/Province	Postal Code	Division	Region	Product ID	Product Name	Sales	Units
0	1	US-2021-103800-CHO-MIL-31000	03-01-2024	30-06-2026	Standard Class	103800	United States	Houston	Texas	77095	Chocolate	Interior	CHO-MIL-31000	Wonka Bar - Milk Chocolate	6.50	2
1	2	US-2021-112326-CHO-TRI-54000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-TRI-54000	Wonka Bar - Triple Dazzle Caramel	7.50	2
2	3	US-2021-112326-CHO-NUT-13000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-NUT-13000	Wonka Bar - Nutty Crunch Surprise	10.47	3
3	4	US-2021-112326-CHO-SCR-58000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-SCR-58000	Wonka Bar - Scrumdiddlyumptious	10.80	3
4	5	US-2021-141817-CHO-TRI-54000	05-01-2024	05-07-2026	Standard Class	141817	United States	Philadelphia	Pennsylvania	19143	Chocolate	Atlantic	CHO-TRI-54000	Wonka Bar - Triple Dazzle Caramel	11.25	3

Figure 1: Dataset Preview

This figure shows the first few records of the Nassau Candy Distributor dataset. It provides an overview of the available columns and the type of information used in the analysis.

Missing Value Analysis

Missing values:

	Missing Values	Missing %
Row ID	0	0.00
Order ID	0	0.00
Order Date	0	0.00
Ship Date	0	0.00
Ship Mode	0	0.00
Customer ID	0	0.00
Country/Region	0	0.00
City	0	0.00
State/Province	0	0.00
Postal Code	0	0.00
Division	0	0.00
Region	0	0.00
Product ID	0	0.00
Product Name	0	0.00
Sales	0	0.00
Units	0	0.00
Gross Profit	0	0.00
Cost	0	0.00

Duplicate rows: 0
Duplicate Row IDs: 0
Duplicate Order IDs: 1645

Figure 2: Missing Value Analysis

This figure shows the missing-value check performed on the dataset. No missing values were found, indicating that the dataset did not require missing-value treatment.

Descriptive Statistics of the Dataset

	count	mean	std	min	25%	50%	75%	max
Sales	10,194.00	13.91	11.34	1.25	7.20	10.80	18.00	260.00
Units	10,194.00	3.79	2.23	1.00	2.00	3.00	5.00	14.00
Gross Profit	10,194.00	9.17	6.64	0.25	4.90	7.47	12.25	130.00
Cost	10,194.00	4.74	5.06	0.60	2.40	3.60	5.70	130.00

```
Negative values:
Sales           0
Units           0
Gross Profit    0
Cost            0
dtype: int64

Zero values:
Sales           0
Units           0
Gross Profit    0
Cost            0
dtype: int64

Maximum absolute Gross Profit difference: 0.0
```

Figure 3: Descriptive Statistics of the Dataset

This figure presents the statistical summary of numerical columns, including count, mean, minimum, maximum, and standard deviation. It helps understand the distribution of sales, units, and other numerical values.

Overall Sales and Profitability Performance

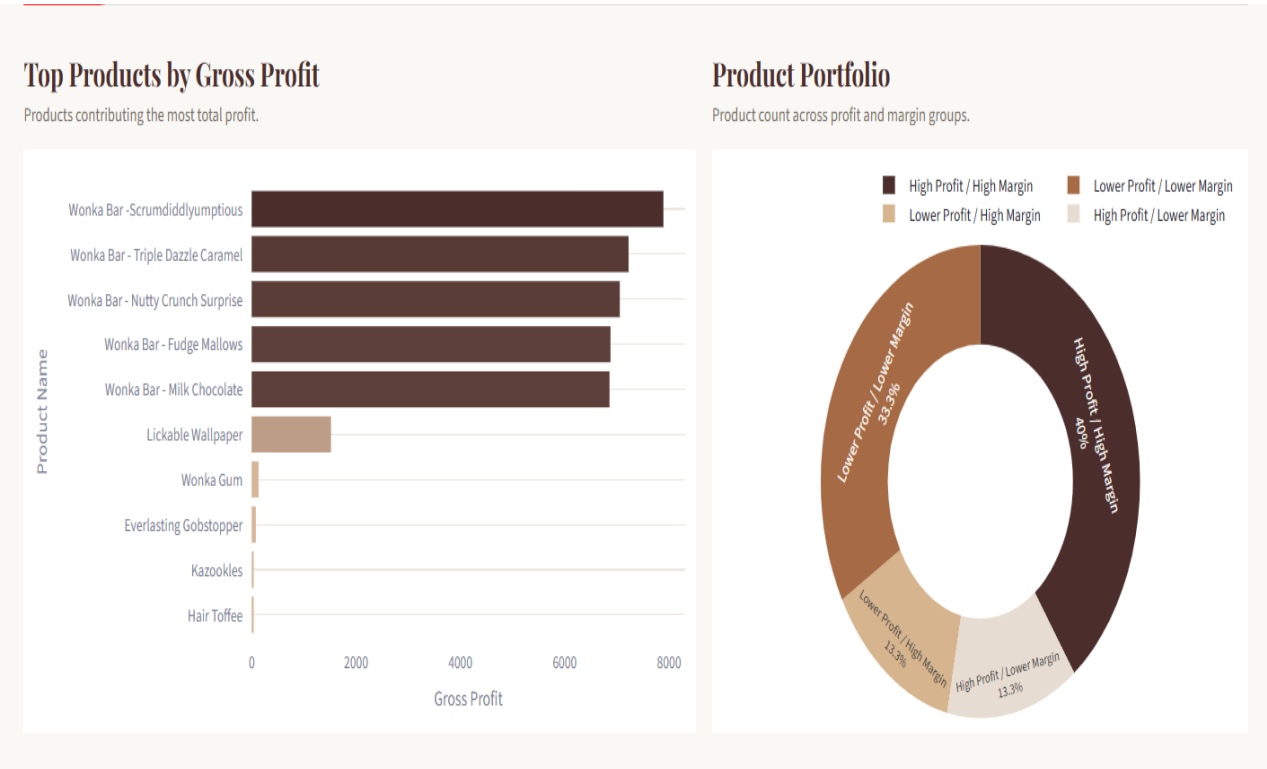


Figure 4: Overall Sales Performance

This figure summarizes the overall financial performance of the business using sales, cost, gross profit, and gross profit margin. It provides a quick overview of product profitability.

Key Performance Indicators (KPI) Dashboard

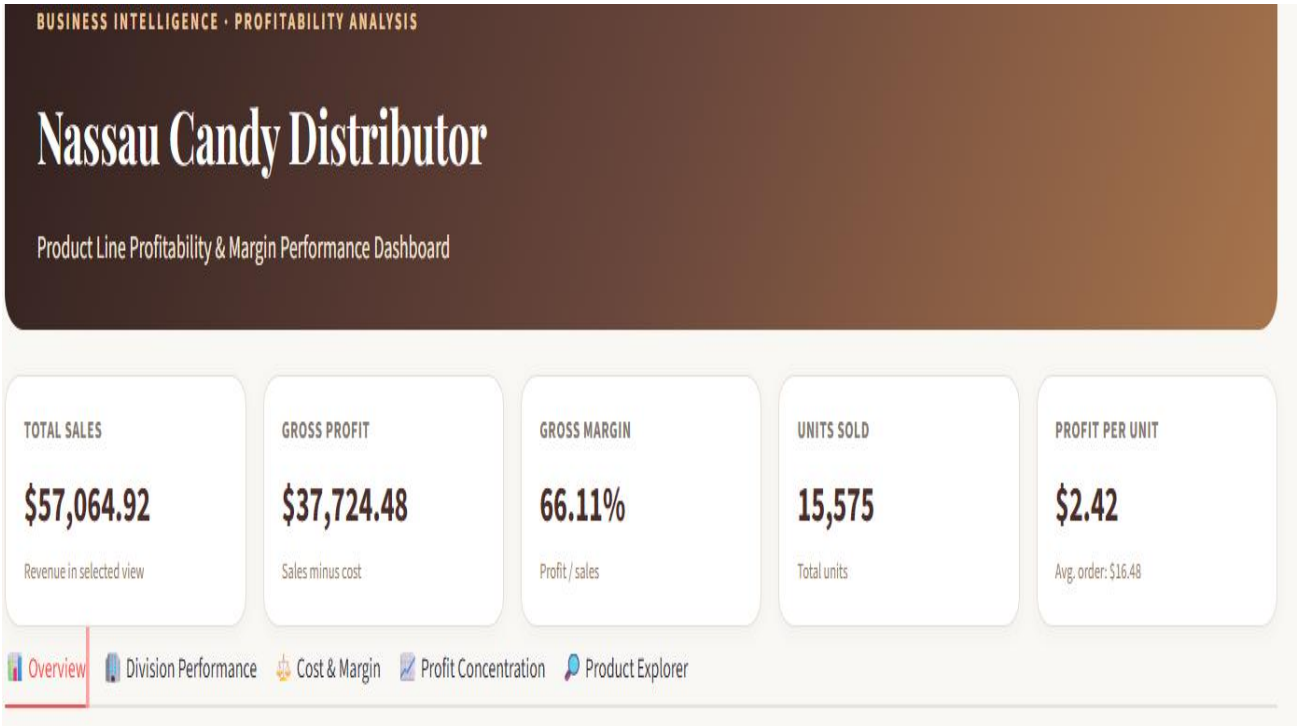


Figure 5: KPI Dashboard

This figure displays the main performance indicators in the dashboard. These KPIs help users quickly understand the overall sales and profitability of the business.

Sales and Gross Profit by Product

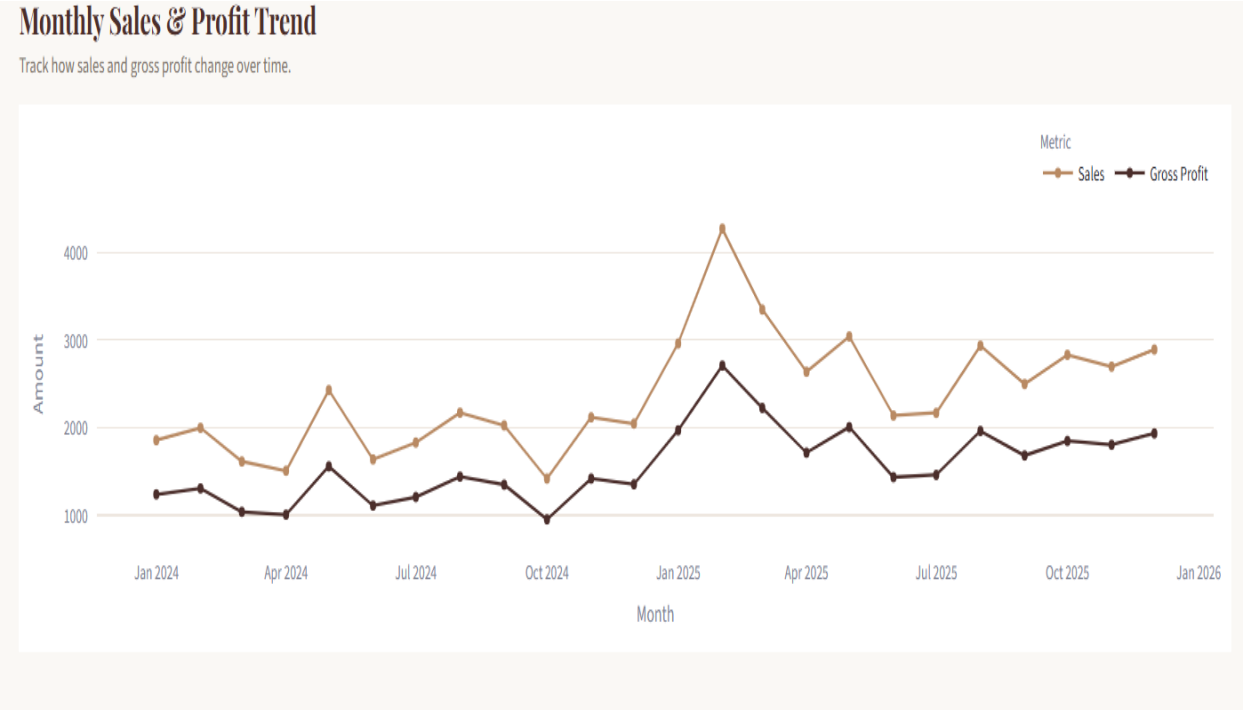


Figure 6: Sales and Gross Profit by Product

This figure compares sales and gross profit for different products. It helps identify products that generate higher revenue and contribute more to total profit.

Division-wise Sales and Profit Comparison

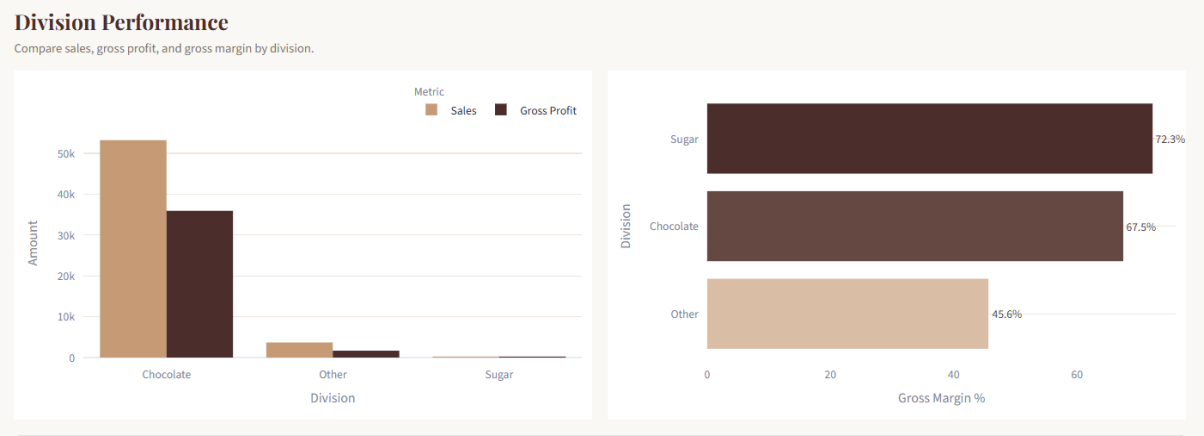


Figure 7: Division-wise Sales and Profit Comparison

This figure compares sales and gross profit across the Chocolate, Other, and Sugar divisions. It helps identify the division that contributes the most to business performance.

Gross Profit Margin by Product

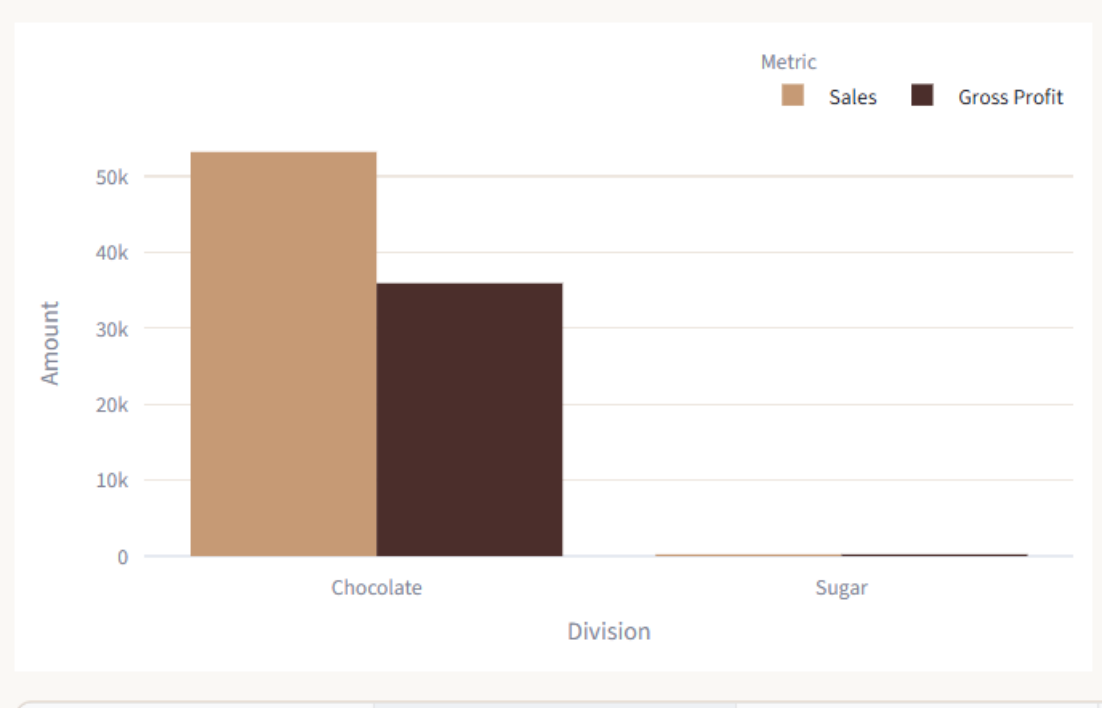


Figure 8: Gross Profit Margin by Product

This figure compares gross profit margins across products. It helps identify products with high margins and products that may require cost or pricing improvements.

Product Cost vs. Sales Analysis

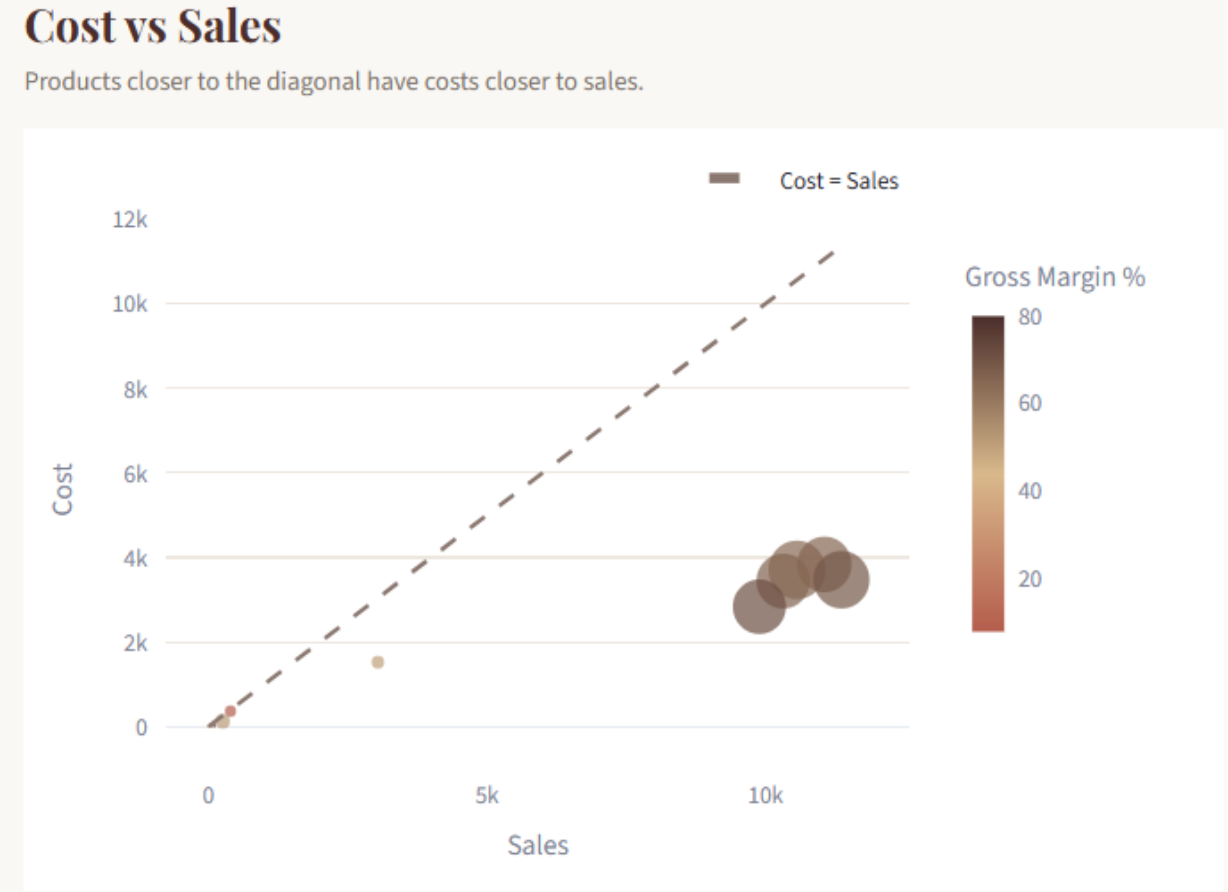


Figure 9: Product Cost vs. Sales Analysis

This figure compares sales revenue with product costs. It helps show how costs relate to sales and supports the evaluation of product profitability.

Product Profitability Ranking

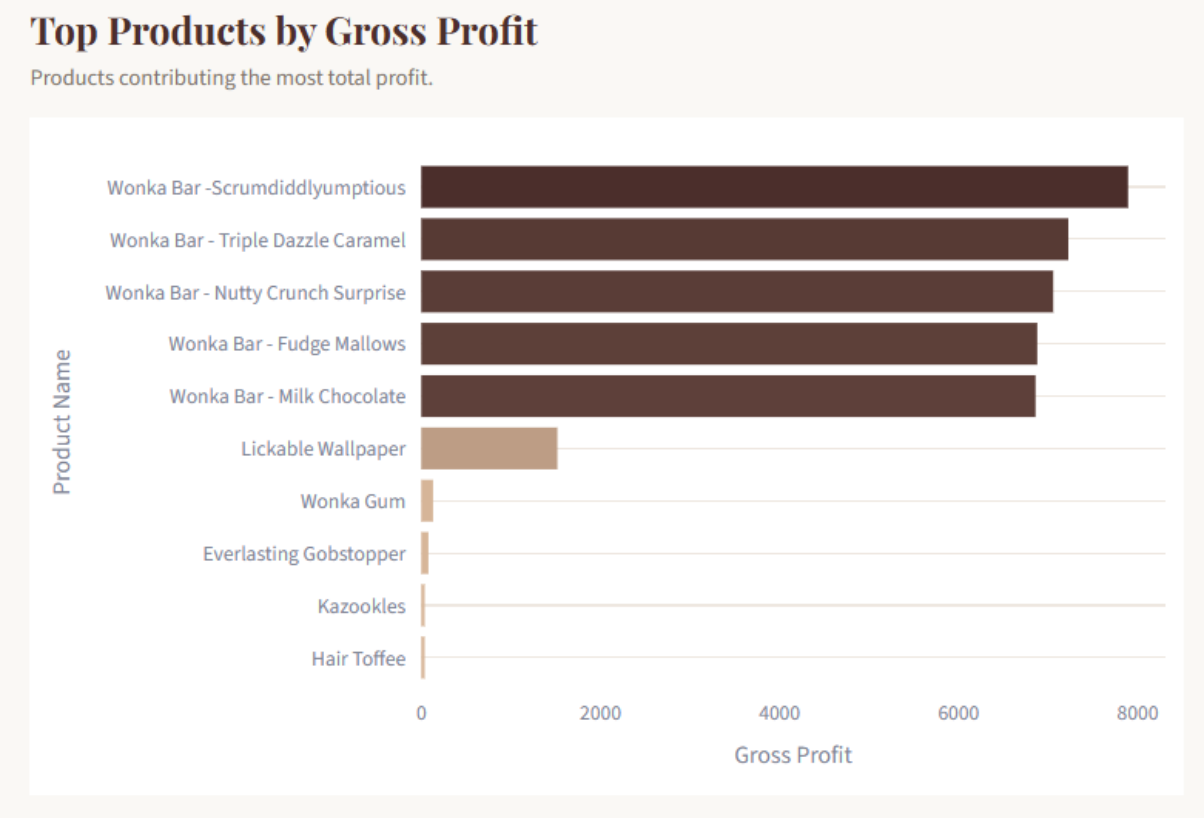


Figure 10: Product Profitability Ranking

This figure ranks products according to gross profit. It highlights the products that contribute the most to overall profit.

Profit Concentration (Pareto) Analysis

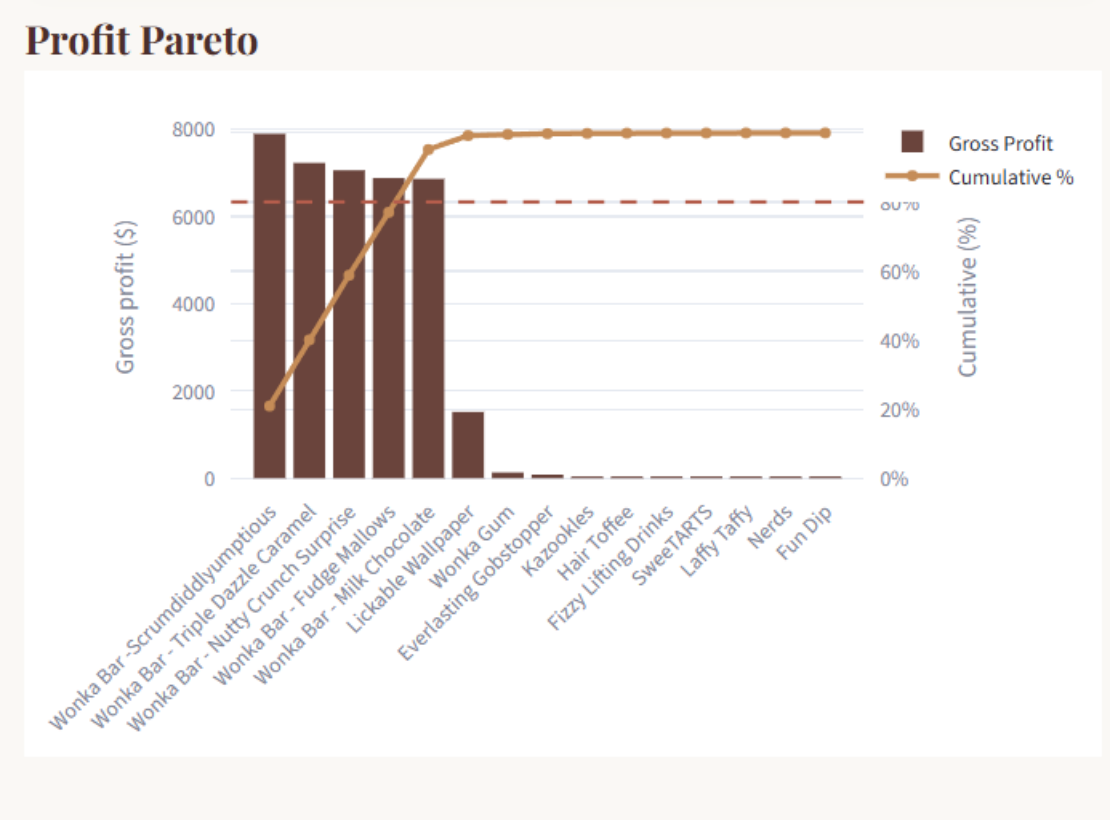


Figure 11: Profit Concentration (Pareto) Analysis

This figure shows how total gross profit is distributed among products. It helps determine whether a small number of products contribute a large share of the total profit.

Interactive Streamlit Profitability Dashboard

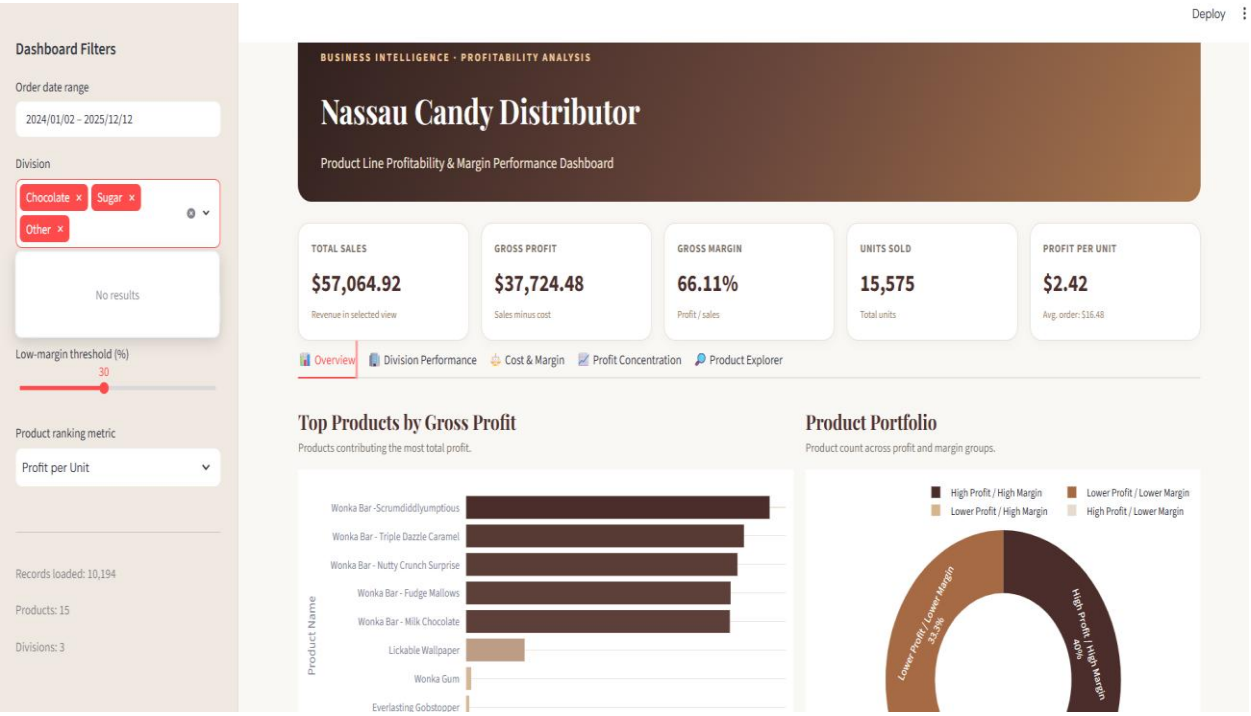


Figure 12: Interactive Streamlit Profitability Dashboard

This figure presents the overall layout of the developed dashboard, including charts, KPI cards, filters, and analytical sections. It demonstrates how the analysis is presented interactively.

Filtered Product and Division Analysis

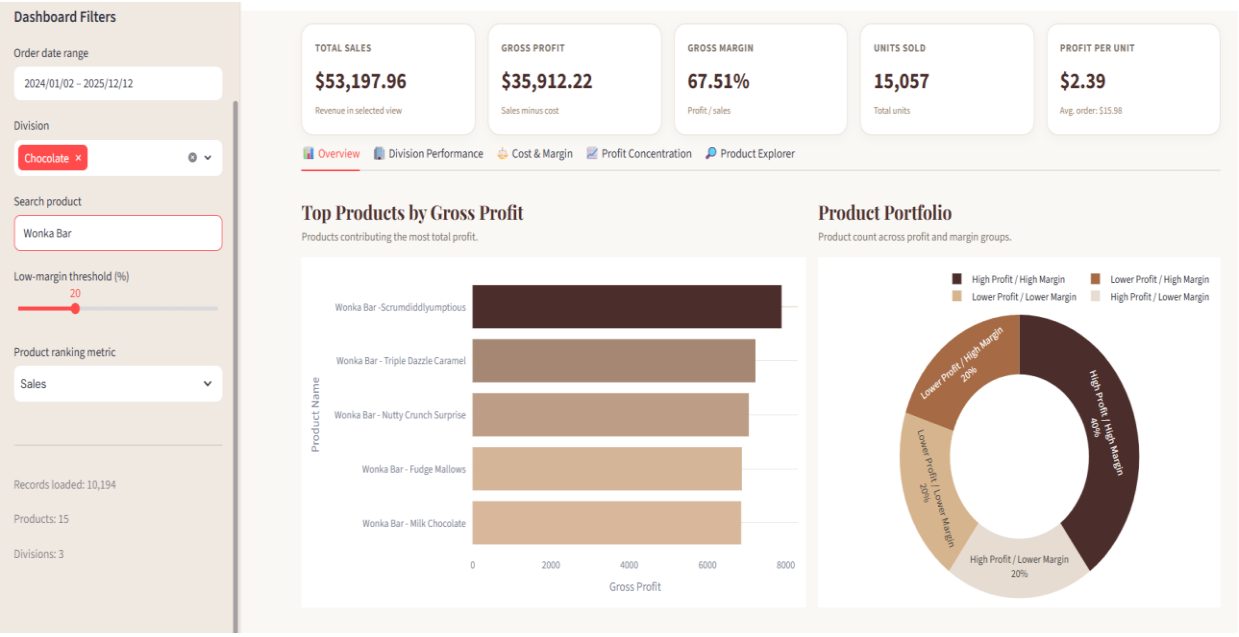


Figure 13: Filtered Product and Division Analysis

This figure shows the dashboard after applying a product or division filter. It demonstrates how users can explore selected categories and compare their profitability.

Executive Summary Dashboard

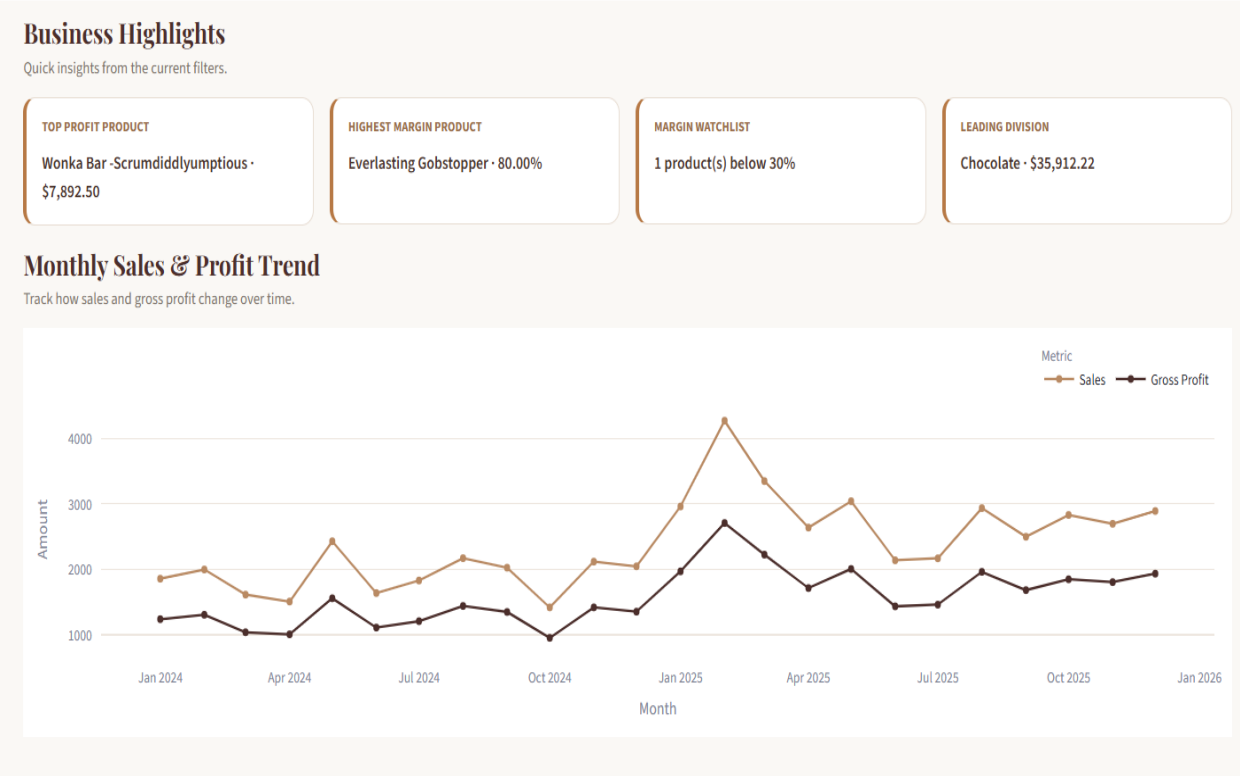


Figure 14: Executive Summary Dashboard

This figure presents the main findings and business insights from the analysis. It helps users understand the important profitability trends and identify areas for improvement.

Results & Findings

The analysis of the Nassau Candy Distributor dataset produced the following results:

- Total Sales: $141,783.63
- Total Cost: $48,340.83
- Gross Profit: $93,442.80
- Gross Profit Margin: 65.91%
- Total Units Sold: 38,654

The Chocolate division contributed the highest sales and gross profit. Wonka Bar – Scrumdiddlyumptious generated the highest gross profit among the listed products, at $19,357.50. Kazookles had a low gross profit margin of 7.69%, indicating a potential area for further investigation. The interactive Streamlit dashboard presents these findings through KPI cards, charts, and filters, making product and division performance easier to compare.

Business Insights

1. Chocolate is the major profit contributor. The Chocolate division generated the largest share of sales and gross profit. Therefore, maintaining product quality, availability, and customer demand within this division is important for the overall business.
2. A few products generate high gross profit. Wonka Bar – Scrumdiddlyumptious, Wonka Bar – Triple Dazzle Caramel, and Wonka Bar – Milk Chocolate were among the leading products by gross profit. These products deserve attention when reviewing product availability and sales performance.
3. High gross profit and high margin are different measures. Wonka Bar – Scrumdiddlyumptious generated the highest gross profit among the listed products, while Wonka Bar – Nutty Crunch Surprise had the highest margin among those five products. Management should consider both measures when evaluating product performance.
4. Kazookles has a low gross profit margin. Its margin of approximately 7.69% is considerably lower than the overall margin of 65.91%. The business could review its selling price, cost structure, and other relevant product expenses to understand the reason for this difference.
5. The Other division needs further investigation. Its gross profit margin of 44.85% was lower than the margins of the Chocolate and Sugar divisions. A detailed product-level review may help identify opportunities to improve profitability.
6. Sales volume alone does not determine profitability. Products should be evaluated using sales, units sold, gross profit, and gross profit margin together. This provides a more complete picture than ranking products only by revenue.
7. Interactive dashboards simplify analysis. The Streamlit dashboard allows users to filter products and divisions and examine results without manually reviewing the complete dataset. This improves the accessibility of the analysis.

These insights describe patterns in the available dataset. Before making actual pricing or operational decisions, the business should also consider demand, competition, fixed expenses, and other relevant costs that may not be represented in the dataset.

Conclusion

The project successfully analyzed product line profitability and margin performance for Nassau Candy Distributor using data analysis and visualization techniques. The dataset was examined using Python and its supporting libraries, and the results were presented through an interactive Streamlit dashboard.

The analysis showed total sales of approximately $141,783.63, total cost of $48,340.83, gross profit of $93,442.80, and an overall gross profit margin of 65.91%. The Chocolate division was the largest contributor to total sales and gross profit. Product-level analysis identified Wonka Bar – Scrumdiddlyumptious as the highest gross-profit product among the products examined, while Kazookles had a comparatively low gross profit margin.

The project demonstrates how Exploratory Data Analysis (EDA), profitability calculations, and interactive visualizations can help transform raw business data into understandable information. The dashboard provides a convenient way to compare products, examine divisions, and identify areas that may need further attention.

Overall, the project provides a useful foundation for understanding product profitability and supporting business decisions. However, the findings are based on the available dataset and should be considered alongside other financial and operational information before being used for actual business decisions.

Future Scope

1. **Sales Forecasting:** Future versions can include forecasting techniques to estimate future sales and help businesses plan product demand.
2. **Advanced Profitability Analysis:** Additional financial information, such as operating expenses, transportation costs, discounts, and marketing expenses, can be included to calculate net profitability more accurately.
3. **Customer Analysis:** Customer-level data can be analyzed to identify purchasing patterns, valuable customers, and opportunities to improve customer retention.
4. **Regional Performance Analysis:** Sales and profitability can be compared across different regions, countries, and cities to identify strong and weak markets.
5. **Pricing Optimization:** Further analysis can investigate how changes in product prices and costs may affect gross profit and profit margins.
6. **Improved Dashboard Features:** The Streamlit dashboard can be enhanced with more filters, downloadable reports, additional visualizations, and automated business summaries.
7. **Machine Learning Integration:** Machine learning techniques could be explored for sales prediction, product grouping, and identifying products with unusual profitability patterns.

These improvements could make the project more useful for detailed financial analysis and long-term business planning.

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